

Seismoelectric conversion generated from water-oil boundary in unsaturated porous media

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Summary

Seismoelectric method is an active geophysical method that measures electromagnetic disturbances associated with the conversion of hydromechanical to electromagnetic energy at macroscopic interfaces characterized by variation in the electrical, hydraulic, and/or mechanical properties. The seismoelectric method has the potential to be used to detect change in saturation during water flushing (injection) of an oil contaminated aquifer and more importantly to track the position of the oil-water encroachment front. The simulated seismoelectric data shows both a strong seismoelectric conversion associated with the oil water interface and a co-seismic signal associated with the arrival of the seismic waves in the second well.

Introduction

We investigate the potential usefulness of the seismoelectric method in the case of water flush / injection of an aquifer contaminated with oil as a potential method to localize the evolution of the oil water encroachment front over time. Until recently the development of the seismoelectric method for aquifer simulations has been slowed down by the lack of model working in two phase flow conditions. Revil and Mahardika (2013) have recently developed a model that can be used to simulate seismoelectric data in unsaturated aquifer conditions. The model is adapted for two-phase flow conditions and a numerical application is shown for water flushing / flooding of a non-aqueous phase liquid (NAPL, oil) contaminated aquifer. Seismoelectric conversions are mostly taking place at the NAPL (oil)/water encroachment front and can be therefore used to remotely track the position of this front. This is not the case for other geophysical methods.

Seismoelectric method (e.g., Ivanov, 1939; Frenkel, 1944; Dupuis and Butler, 2006) and the coupled formulation of hydromechanical and electromagnetic disturbances on unsaturated media in general have broad applications for the assessment of water resources and remediation. Dupuis et al. (2007) has demonstrated the feasibility to measure seismoelectric signal for vadose zone applications. Solutes and NAPLs (and also dense-NAPLs / DNAPLs) in the vadose zone can be moved using electroosmosis (Han et al., 2004), and clayey soils can be dewatered in civil engineering operation (Bjerrum, 1967).

Theory

We consider below an isotropic porous material with connected pores (an idealized soil with a representative elementary volume). The surface of the minerals in contact with water is considered to be negatively charged at near neutral pH values (5 to 8). There is therefore an excess of (positive) charges in the pore space of the soil in an electrical double layer. In the following, we use the subscript *a* to describe the properties of air, *o* to describe the properties of NAPL / oil (air and NAPL act as the non-wetting phase fluids), and subscripts *w* and *s* for the water and solid phases, respectively. Water is assumed to be the wetting phase.

Assuming an isotropic and fully-saturated condition, a set of governing equations for coupled electromagnetic and acoustics in heterogeneous poroelastic media is proposed by Pride (1994) based on elastic wave propagation described by Biot's theory (Biot, 1956) Considering the frequency dependence of material properties ($e^{-i\omega t}$ where ω is angular frequency and $i^2 = -1$), the coupled formulation of poroelastic and electrical disturbances is extended for unsaturated porous media which are given as (Revil & Mahardika, 2013)

$$-\omega^2 \rho_o^s \mathbf{u} + \theta_o \nabla p_w = \nabla \cdot \bar{\bar{T}}_{eff} \quad (1)$$

$$\bar{\bar{T}} + p_a \bar{\bar{I}} + \alpha_w (p_w - p_a) \bar{\bar{I}} = \bar{\bar{T}}_{eff} \quad (2)$$

$$\bar{\bar{T}}_{eff} = \lambda_u (\nabla \cdot \mathbf{u}) \bar{\bar{I}} + G (\nabla \mathbf{u} + \nabla \mathbf{u}^T) \quad (3)$$

$$\left(\frac{1}{M} + \frac{\partial \theta}{\partial p_w} \right) (p_a - p_w) = \alpha_w \nabla \cdot \mathbf{u} + \nabla \cdot \left[\frac{k^*(\omega)}{i\omega \eta_w} (\nabla p_w - \omega^2 \rho_w s_w \mathbf{u}) \right] \quad (4)$$

$$\nabla \cdot \sigma \nabla \psi = \nabla \cdot \mathbf{j}_S \quad (5)$$

$$\mathbf{j}_S = \frac{\hat{Q}_V}{s_w} \frac{k^*(\omega)}{\eta_w} (\nabla p_w - \omega^2 \rho_w s_w \mathbf{u}) \quad (6)$$

where ρ_o^s is the apparent mass density of the solid phase, $\mathbf{u}$ is the solid displacement, p_a and p_w denote the average air and water pressures (in Pa), θ_o is the hydro-mechanical coupling coefficient, $\bar{\bar{T}}_{eff}$ is the effective stress tensor, $\bar{\bar{T}}$ is the background stress tensor, $\bar{\bar{I}}$ is an identity matrix, α_w is Biot coefficient, λ_u is the undrained Lamé parameter, G is the shear modulus, M is Biot modulus, ρ_w denoting water density, $k^*(\omega)$ is the complex-valued apparent permeability, η_w is water viscosity, and $\theta = \phi s_w$ is the water content with ϕ and s_w denoting porosity and water saturation respectively. Full definition of ρ_o^s , θ_o ,

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and $k^*(\omega)$ are described extensively in Revil and Mahardika (2013). The electromagnetic response was solved using equation. 5 and 6, where; ψ is the electrostatic potential ($\mathbf{E} = -\nabla\psi$ in the quasi-static limit of the Maxwell equations which $\mathbf{E}$ is the electric field), σ is the electrical conductivity of the medium, $\mathbf{j}_s$ is the source current density of electrokinetic nature, and $\hat{Q}_V$ is the excess of charge on the diffuse layer per unit pore volume (C/m^3).

The electrokinetic coupling in the seismoelectric response is the result of the relative displacement of the pore water with respect to the solid phase, explained by equation. 4 to 6. The drag of the charge density contained in the pore water is responsible for a net source current density in a framework attached to the solid phase (Leroy and Revil, 2004). The electromagnetic disturbances are considered in the quasi-static limit of the Maxwell equations as in the self-potential studies (Suski et al., 2006; Jardani et al., 2007). This assumption is considered since the observation domain is less than 1 km from the receivers' arrays (dipole antennas, and non-polarizing electrodes). Because of these assumption, the time required by the electromagnetic disturbances to diffuse between the aquifer and receivers can be neglected.

Model

We consider two wells located in a heterogeneous aquifer. Well B is located 250 m away from Well A. The region is initially saturated with NAPL (oil). Water is injected in Well A and NAPL is produced in Well B. We produce fluid saturation evolution profiles for a given porosity and fluid permeability input using the multiphase fluid flow equations. The description of the corresponding porosity and permeability distribution generator, the water flooding numerical model, and its application to the unsaturated condition is described extensively in Revil and Mahardika (2013) and will not be repeated here. We use these inputs as foundation for the seismoelectric model together with material properties values given in Table 1, to create p-wave velocity and electrical conductivity distribution. An example of these quantities after 400 days of injection is given in Figure 1. A total of three saturation profiles are going to be used in this study: map at day 0 (before water injection), at day 400, and day 800 after the injection has started (in the following sections, we named these fluid saturation profiles as T1 T2, and T3 respectively).

Table 1. Material properties used in the model.

Parameter	Value	Units
ρ_s	2650	kg m ⁻³
ρ_w	1000	kg m ⁻³

ρ_o	900	kg m ⁻³
K_s	36.5	GPa
K_{fr}	5.8	GPa
G	3.96	GPa
K_w	2.25	GPa
K_o	1.5	GPa
η_w	1.10^{-3}	Pa s
η_o	5.10^{-2}	Pa.s

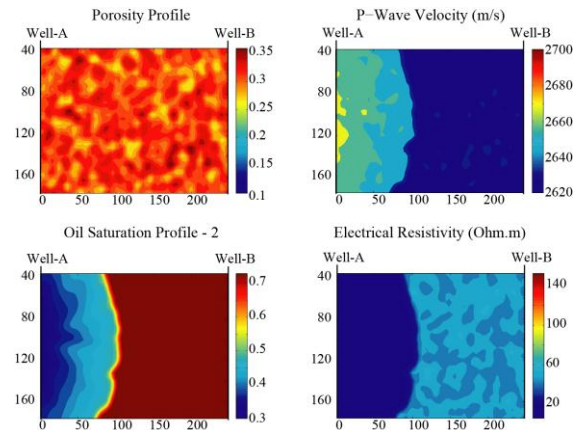


Figure 1: Porosity, oil saturation, p-wave velocity and electrical resistivity map of NAPL (oil) contaminated aquifer between two wells after 400 days (T2) of water-injection.

The geometry for the synthetic seismic and electric signal propagation is shown in Figure 1. An impulse-like source time function of a seismic source is located in Well A (which also act as a seismic source well) at So(0 m, 104 m). We set the seismic source has magnitude of $M = 1.0 \times 10^3$ N m, and dominant frequency $f_c = 100$ Hz. The signal receivers, composed of 24 pairs of seismic stations and electrodes, are vertically arranged at Well-B (noted as R1 to R24). The separation between receivers is 4 m. This receiver arrangement mimics the acquisition that would be obtained with triaxial geophones and dipole antennas in crosswell acquisition settings. The reference position, O(0,0), is located at the upper left corner of this domain.

We compute the poroelastic and electric properties distribution given the porosity, fluid permeability, and each saturation profiles, then we solve for the displacement of the solid phase and the pore fluid pressure. Finally, we compute the electrical potential by solving the Poisson equation coupled to the poroelastodynamic problem for frequency range of 5-500 Hz. This range is valid for our problem since the seismic wave and the associated electrical field operates in the same frequency range (Garambois and Dietrich, 2001). We then compute the inverse fast Fourier transform (FFT⁻¹) to get the time series of the seismic displacements u_x and u_z , and the time series of the electric potential response ψ (see Jardani et al., 2010).

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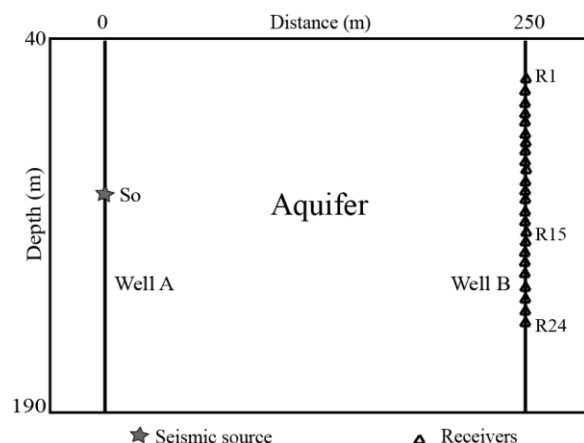


Figure 2: Geometry of the seismic and electric propagation model. Injector Well-A, which is also used for shooting well, located at position $x = 0$ m. Production and recording Well-B is located at $x = 250$ m

Results

We create a set of seismogram depicting horizontal displacements of the solid phase and electrograms from the associated electric potential signals collected at the 24 channel receivers on Well-B shown in Figure 3. In particular, these seismogram and electrogram are computed by utilizing fluid saturation condition after 400 days of water injection (profile T2). The seismic displacements are created from the isotropic-mechanical source which allows only for generation of p-wave. Here, we show that the seismoelectric conversion signal (SE) is uniquely created from the saturation contrast as it is indicated by the zero amplitudes of electric potential data before and after the seismic wave traverse the saturation barrier at 31–47 ms range. Comparing this with the seismic counterparts, we show p-wave arrival in the seismogram and two important signals in the electrograms. In the electrograms we collect the Type I seismoelectric signal denotes electromagnetic disturbances generated at saturation contrast which we denote as seismoelectric conversions (SE), and the Type II signal which corresponds to the co-seismic electromagnetic signals (denotes as CS). In general, signal SE occurs when seismic wave meet any electrical and mechanical heterogeneity in the aquifer (in this case the oil-water encroachment front), while signal CS is an electric field generation that is coupled to the p-wave propagation in porous media (extensive descriptions on seismoelectric signal Type I and II are given in Haines (2004), and reference therein). Since seismoelectric conversion signal in particular is unique to the electric data, the utilization its amplitude at seismoelectric arrival time t_{SE} and also information from co-seismic arrival time t_{CS} from the 24 channels receivers may be useful for the inverse calculation of the water-oil boundary shape via current density of

electrokinetic nature inside the modeling domain (Jardani et al, 2007; Minsley et al, 2007). The inverse calculation, however, is not in the scope of the current paper.

Next, we show complete evolutions of seismic and electric potential time series recorded at receiver-15 for water saturation profiles at day 0 (before water injection), at 400 and 800 days after water injection in Figure 4. Strong seismoelectric conversion (SE) signals occur after the water injection was started at profile T2 and T3, and arrives later as the water front approaching Well B. The minimum trough arrival times of the seismoelectric conversion signal for T2 and T3 are $t_{SE} = 36$ ms and $t_{SE} = 56$ ms respectively. These arrival time variations are again show the existence conversion mechanism at the interface between low-partial saturation zone and high-partial saturation zone on each T2 to T3. In profile T1, we do see seismoelectric conversion signals which are generated from the porosity variation of Figure 1 (shown inside the small marker box on Figure 4). These signals however are relatively small compared to the main conversion effects (SE) and physically not related to the water front movement. Therefore, we will not analyze these arrivals on the inversion section. Co-seismic signal (CS) however does occur on this profile and also on the T2 and T3 settings.

Conclusions

From this study we have developed a framework for the forward modeling of seismic and electrical data associated with a seismoelectric conversion due to a saturation front. The propagation of the seismic waves and electrical signals are done with the finite element approach in the frequency domain. The forward model couples Biot theory with an electrokinetic model based on the excess of electrical charges per unit pore water. The model can be easily modified to account for two-phase flow conditions for applications to remediation problems of NAPL/DNAPL plumes, relevant to the early contamination problem of a clayey sand aquifer. We have shown that the oil water encroachment front can be remotely detected using cross-well SE conversions. Therefore, this opens the door to a broad range of applications to monitor and image remotely changes in the water saturation in the vadoze zone and the monitoring of NAPLs and DNAPLs plumes

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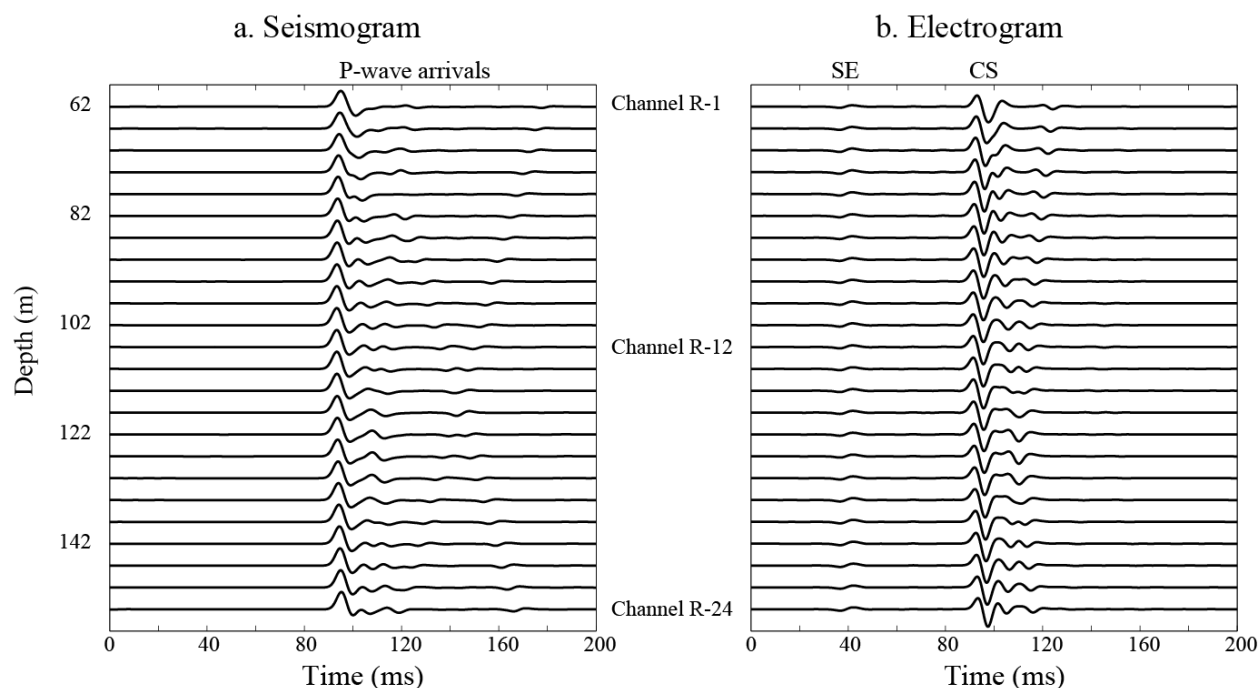


Figure 3: Twenty-four channels seismogram and electrograms from implementing saturation profile T2. (a) The seismograms reconstructed by the geophones show the p-wave propagation from the seismic source in Well A to recording Well B. (b). The electrograms show the coseismic electric potential field associated with the p-wave (CS) and the seismoelectric conversions (SE) with smaller amplitudes and same arrival times.

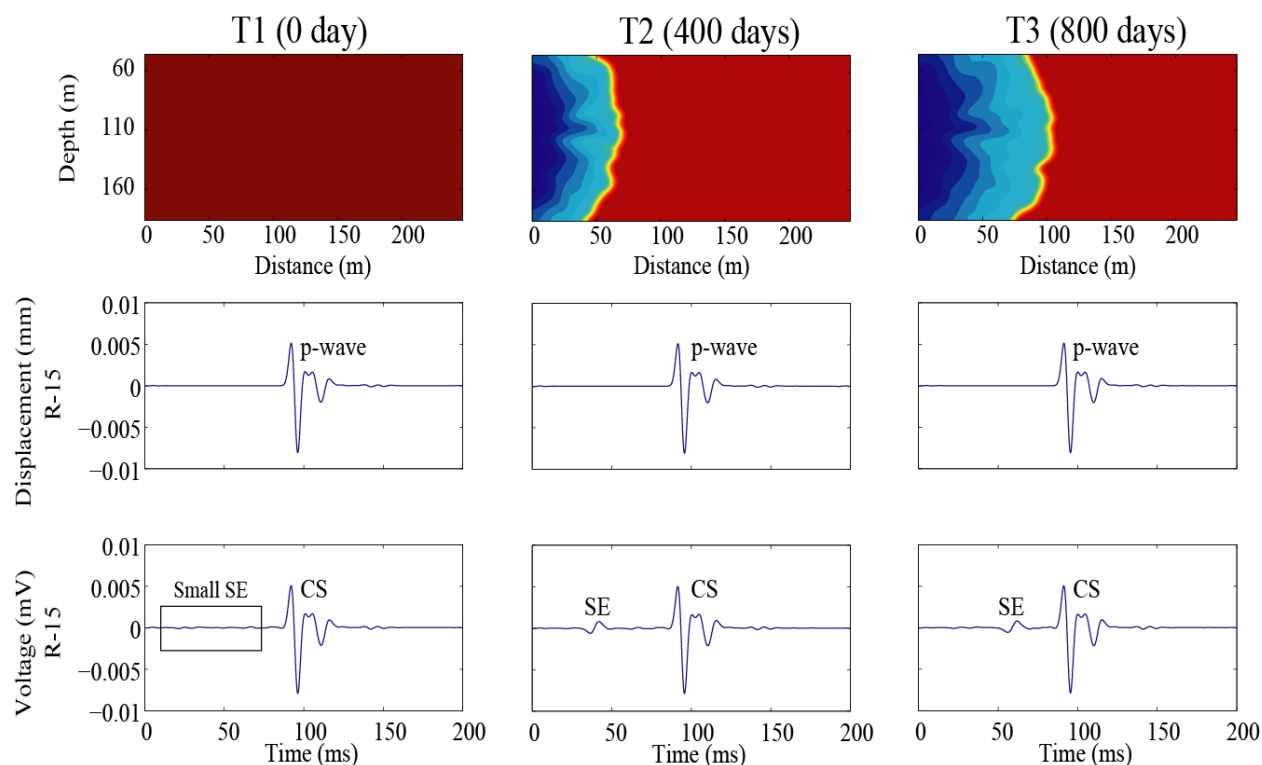


Figure 4: Evolution of the seismic displacement and the associated electric potential time series from receiver R15 due to changes in the position of the oil-water encroachment front during snapshots T1 to T3. The p-wave arrivals in the seismic time series is well identified, as well as the seismoelectric conversions (SE) at interfaces (Type I electrical signal), and CS: Co-seismic P-wave (Type II electrical signal) on the electrical counterpart.

EDITED REFERENCES

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